introduce the preflight cache, which is a browser optimization that helps limit the number of preflight requests that are made.

4.1 What is a preflight request?

Let's think about a preflight request in the context of the ATM example from chapter 3. Banks sometimes put their ATMs inside a room behind a locked door. The door can only be unlocked by swiping your ATM card (or if a kind person lets you in, but let's ignore that for now). Once you're inside, you can walk up to the ATM and withdraw money. The simple act of swiping your card to unlock the door doesn't automatically give you money, but it's a quick check to verify that you have permission to use the ATM.

In a similar fashion, a preflight request asks for the server's permission to send the request. The preflight isn't the request itself. Instead, it contains metadata about it, such as which HTTP method is used and if the client added additional request headers. The server inspects this metadata to decide whether the browser is allowed to send the request.

By asking for permission before making the request, the preflight introduces an additional processing step to CORS. Let's dig deeper into how this new step fits into your existing understanding of CORS.

4.1.1 Lifecycle of a preflight request

Chapter 3 framed CORS requests in the context of a conversation between the browser and the server. The preflight augments this conversation with additional dialogue, as shown in figure 4.1. This conversation is a bit longer than the conversation from chapter 3. It adds the first two lines, where the browser asks the server for permission to use the `DELETE` method. These two lines are the preflight request, while the last two lines are the CORS request.

Figure 4.2 expands figure 3.12 in chapter 3 to show how the preflight request fits into the lifecycle of a CORS request. The browser uses the server's response to the preflight to determine if the request can be made. If the server grants the right permissions on the preflight response, the browser sends the request to the server. The server can also decide not to approve the request, in which case the browser will return an error to the client, and the request will never be sent.

Now that you have a sense of what a preflight request is, let's discuss why it exists in the first place.

4.1.2 Why does the preflight request exist?

The concept of a preflight was introduced to allow cross-origin requests to be made without breaking existing servers that depend on the browser's same-origin policy. If the preflight hits a server that is CORS-enabled, the server knows what a preflight request is and can respond appropriately. But if the preflight hits a server that doesn't know or doesn't care about CORS, the server won't send the correct preflight response, and the actual request will never be sent. The preflight protects unsuspecting servers from receiving cross-origin requests they may not want.